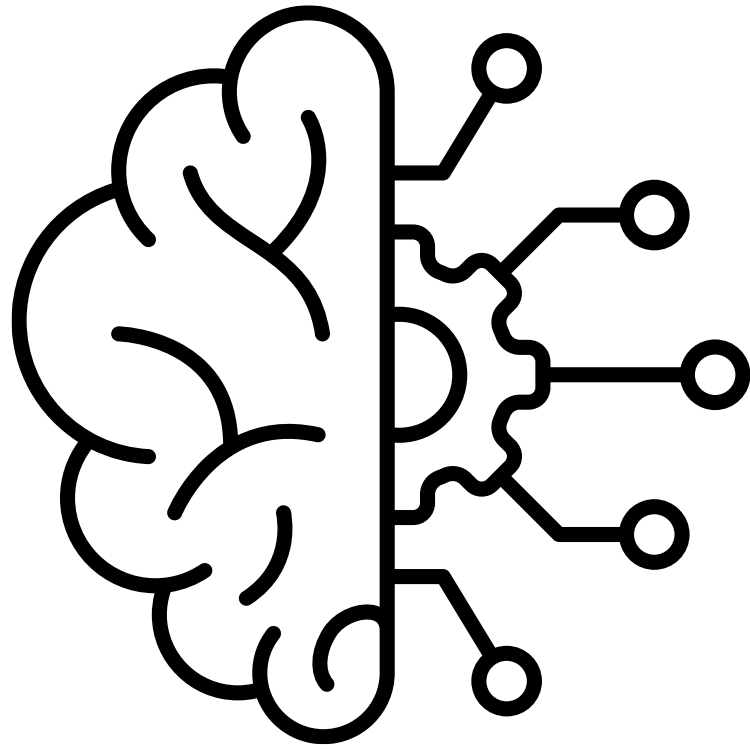




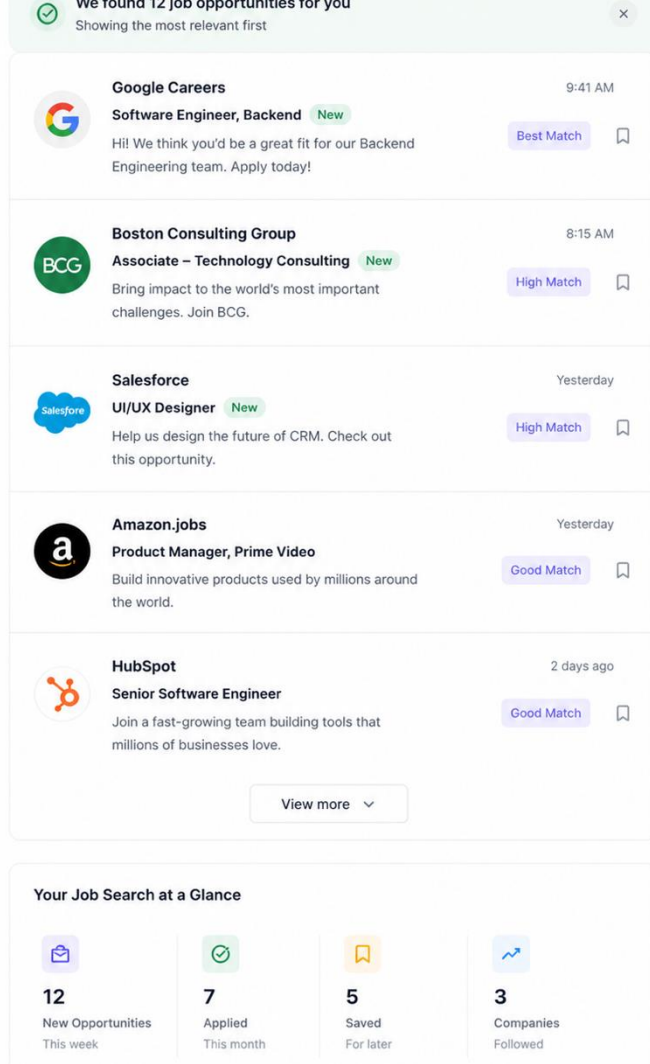
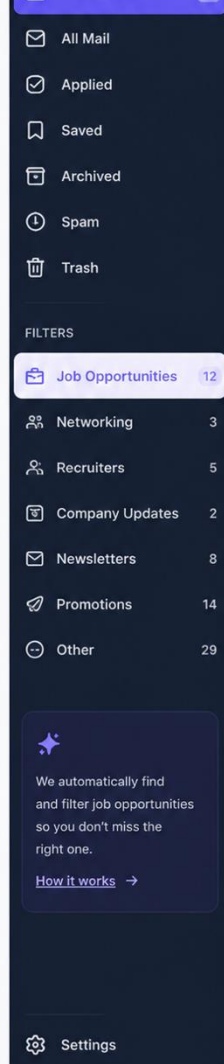
Building AI Agents for the Humanities

Session 3: RAG

Yadira Lizama Mué, PhD

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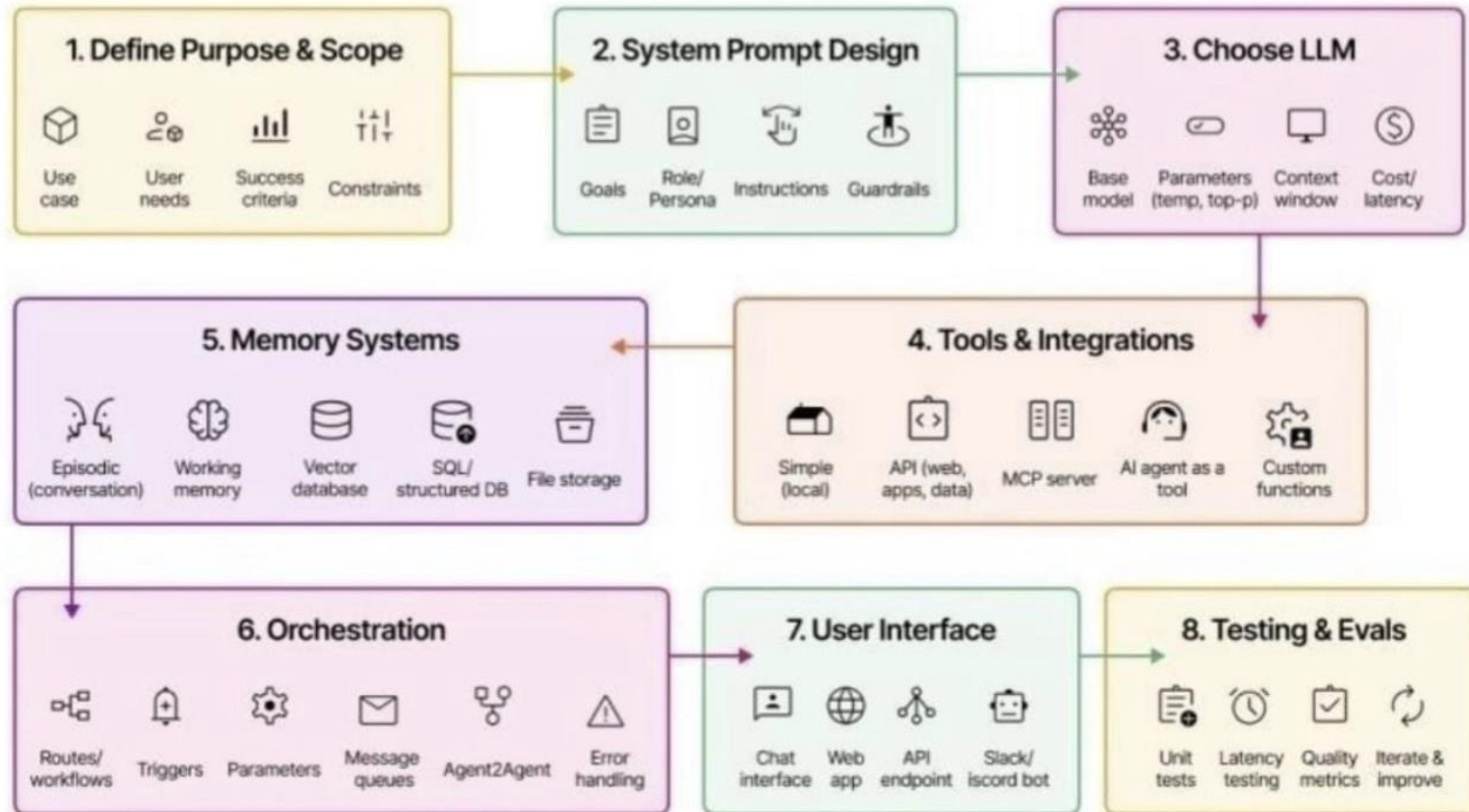




Design Challenge

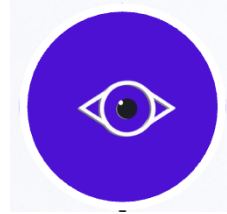
- Your agent must include:
 - **Goal:** What is the agent trying to accomplish?
 - **Prompt(s):** What instructions guide the agent?
 - **Actions:** What is the sequence of actions or decisions?
 - **Tools:** What tools or data sources does it use (e.g., search, APIs, documents)?
 - **Human Role:** Where does human input or judgment matter?

AI Agent - Components



Again... what is an AI Agent?

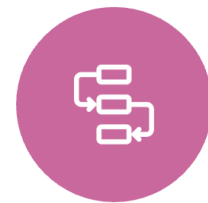
An **AI agent** is a system that:



Perceives its environment (input/data/trigger)



Decides what to do (reasoning/planning)



Acts to achieve a goal (takes actions, often over time)



Adapts based on feedback (optional but common)

New Age of Generative AI

- Massive pre-training on diverse, heterogeneous datasets.
- CoT Prompting techniques for reasoning
- Agentic AI architectures where models generate but also reason, plan and act.
- Multi-agent systems collaborating toward complex goals
- Multimodal generation across text, vision, and audio.
- Improvement in GPU infrastructure and efficient models.

“What started as data mimicry has evolved into intelligent, multimodal agents capable of augmenting and automating human thought itself” (Kar 2026)

Retrieval Systems

- **1970s – 2000s:** Term frequency-inverse document frequency (TF-IDF), Best Matching 25 (BM25); sparse, keyword-based, and focused on matching exact terms.
- **2020:** Dense passage retrieval (DPR); introduced dense embeddings to match questions and documents semantically.
- **2021:** Hybrid retrieval: sparse (BM25) + dense (DPR)
- **2020 – 2022:** RAG; tight integration of retrieval + generation models.
- **2023 +:** Dynamic, reasoning-driven retrieval embedded inside LLM workflows.

Goal: Given a query, find the most relevant existing information

Tokenization

- Break down the text into discrete units that the model can understand and process. Models like GPT-3, GPT-4, and Llama do not operate directly on raw text; they need sequences of tokens.
- The number of tokens determines:
 - Cost (API's pricing is per token)
 - Context window size (how much info the model can see at once)
 - Model behaviour (GPT-4 Turbo supports 128K tokens maximum)

"This is tokenizing."

Character Level

[T] [h] [i] [s] [i] [s] [t] [o] [k] [e] [n] [i] [z] [i] [n] [g] [.]

Word Level

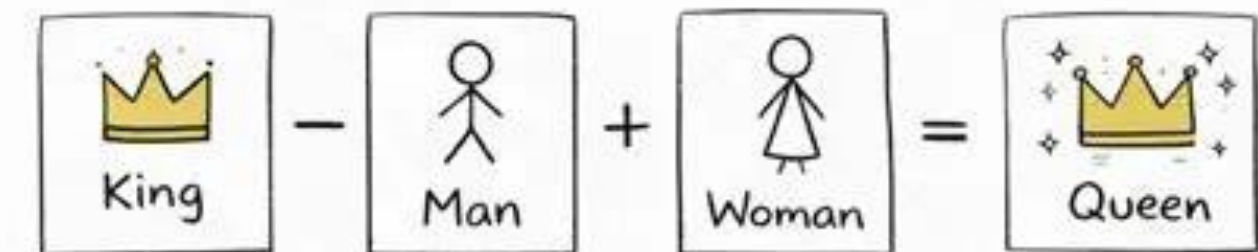
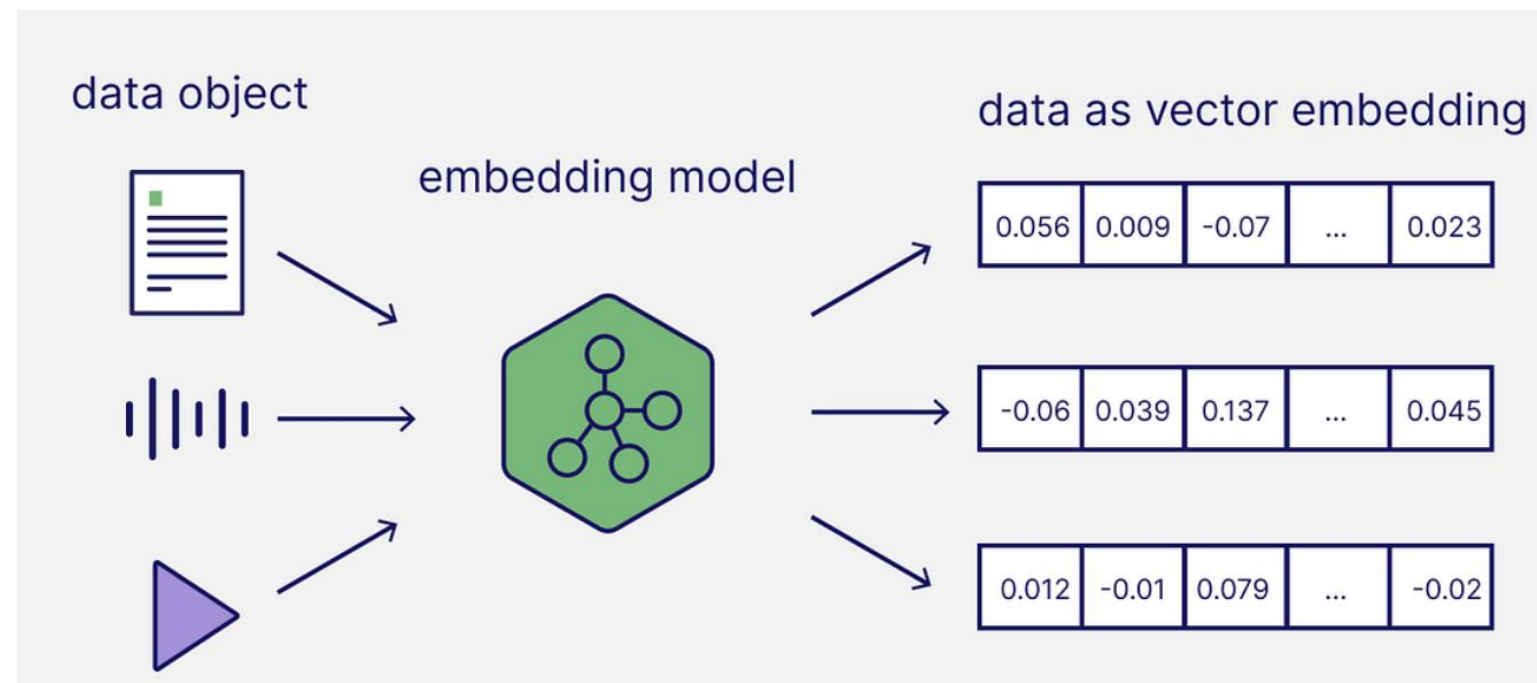
[This] [is] [tokenizing] [.]

Subword Level

[This] [is] [token] [izing] [.]

Embeddings

An **embedding** is a way to represent words, sentences, documents, images, and audio as a dense numerical vector (lists of numbers) that captures their meaning (similar inputs are closer in the vector space).



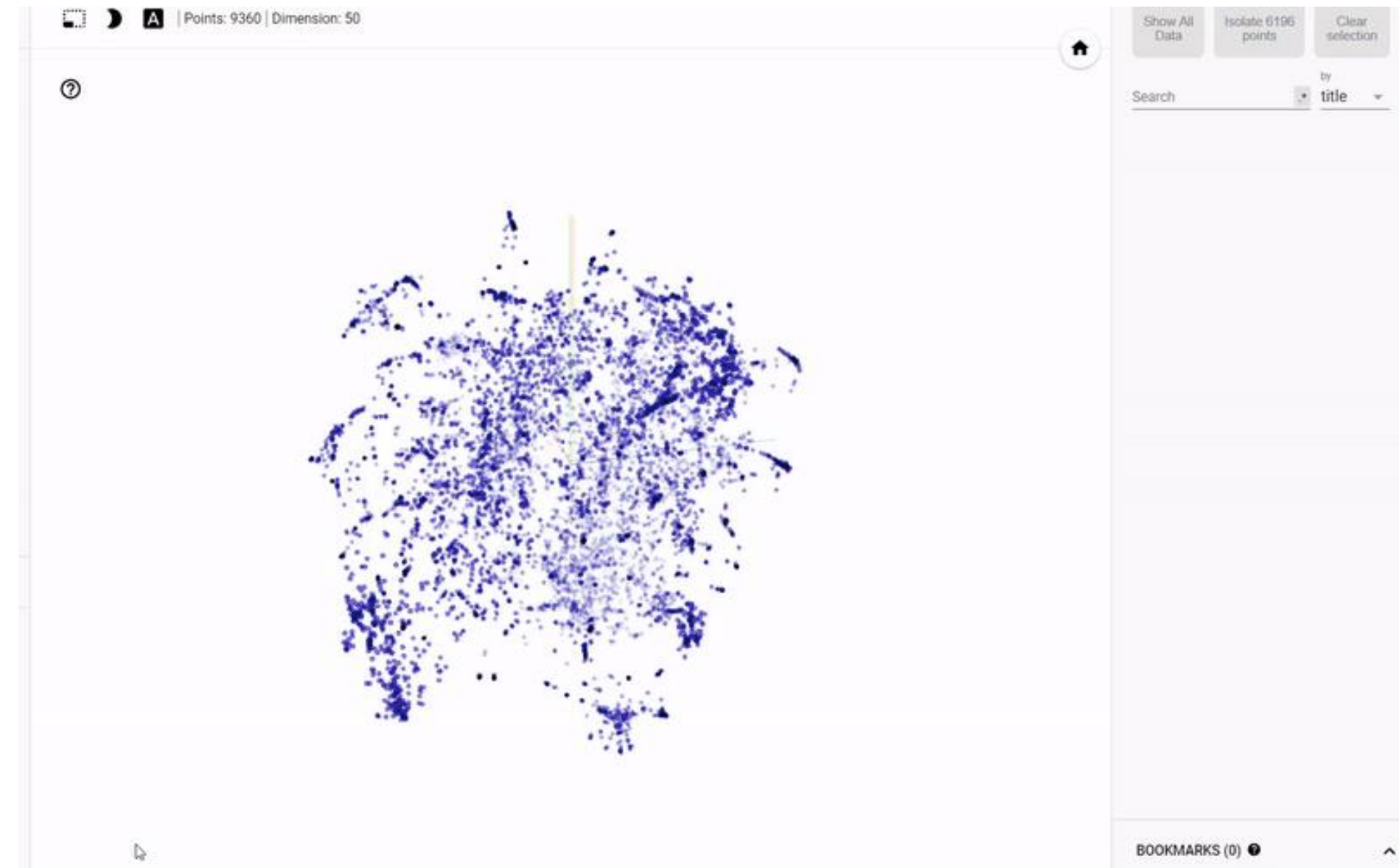
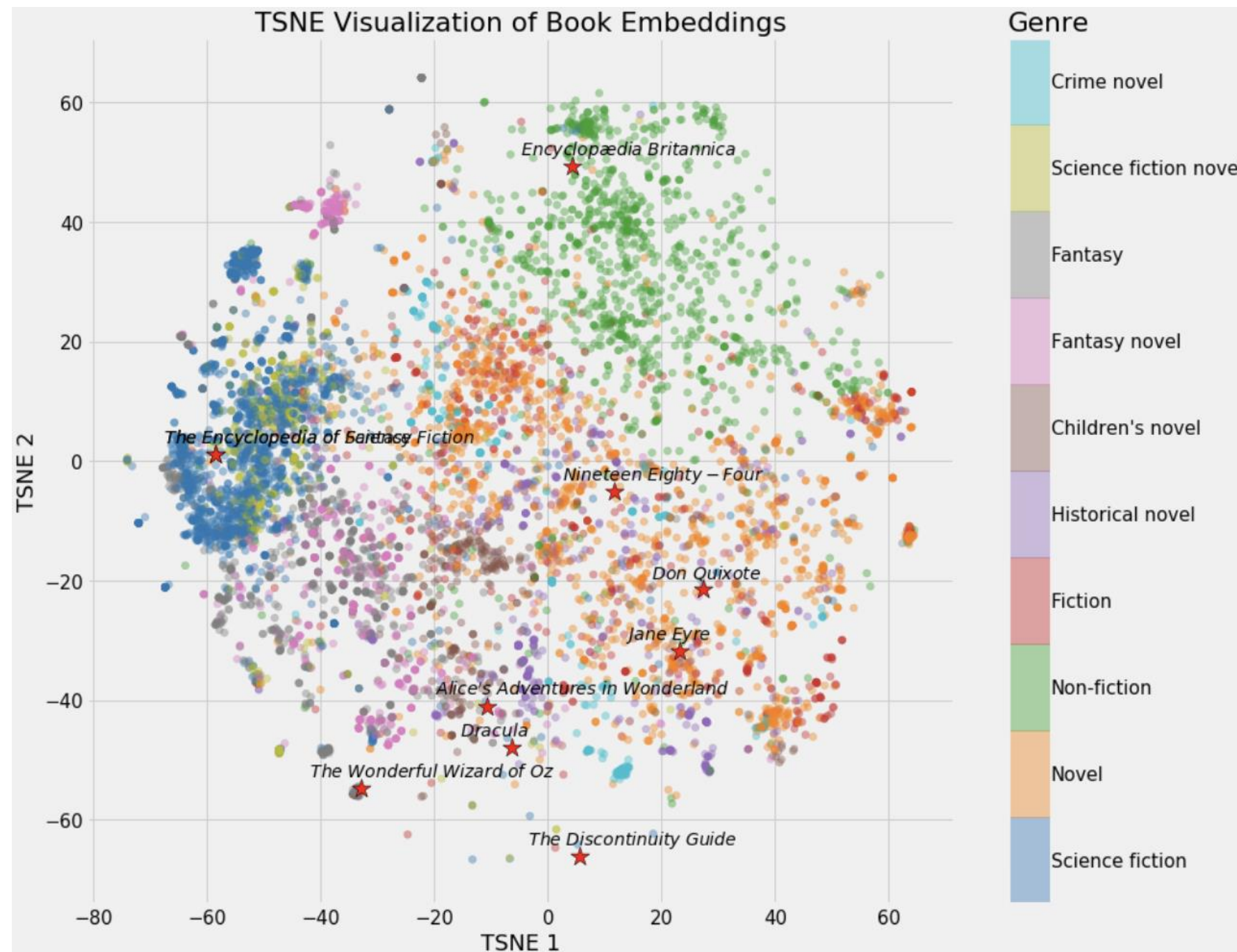
Math on meaning

Paris - France + Italy = Rome

Walking - Walk + Swim = Swimming

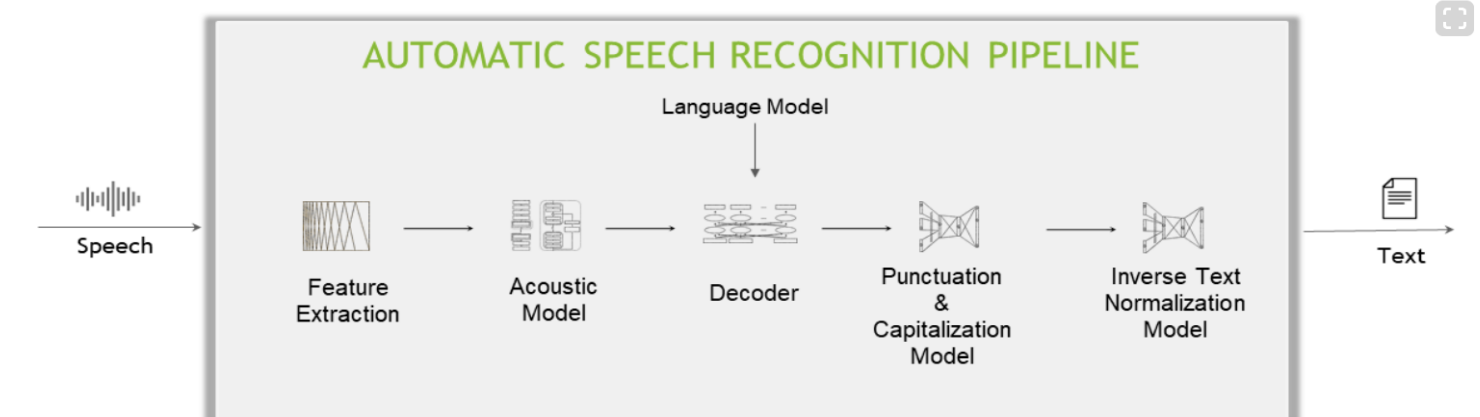
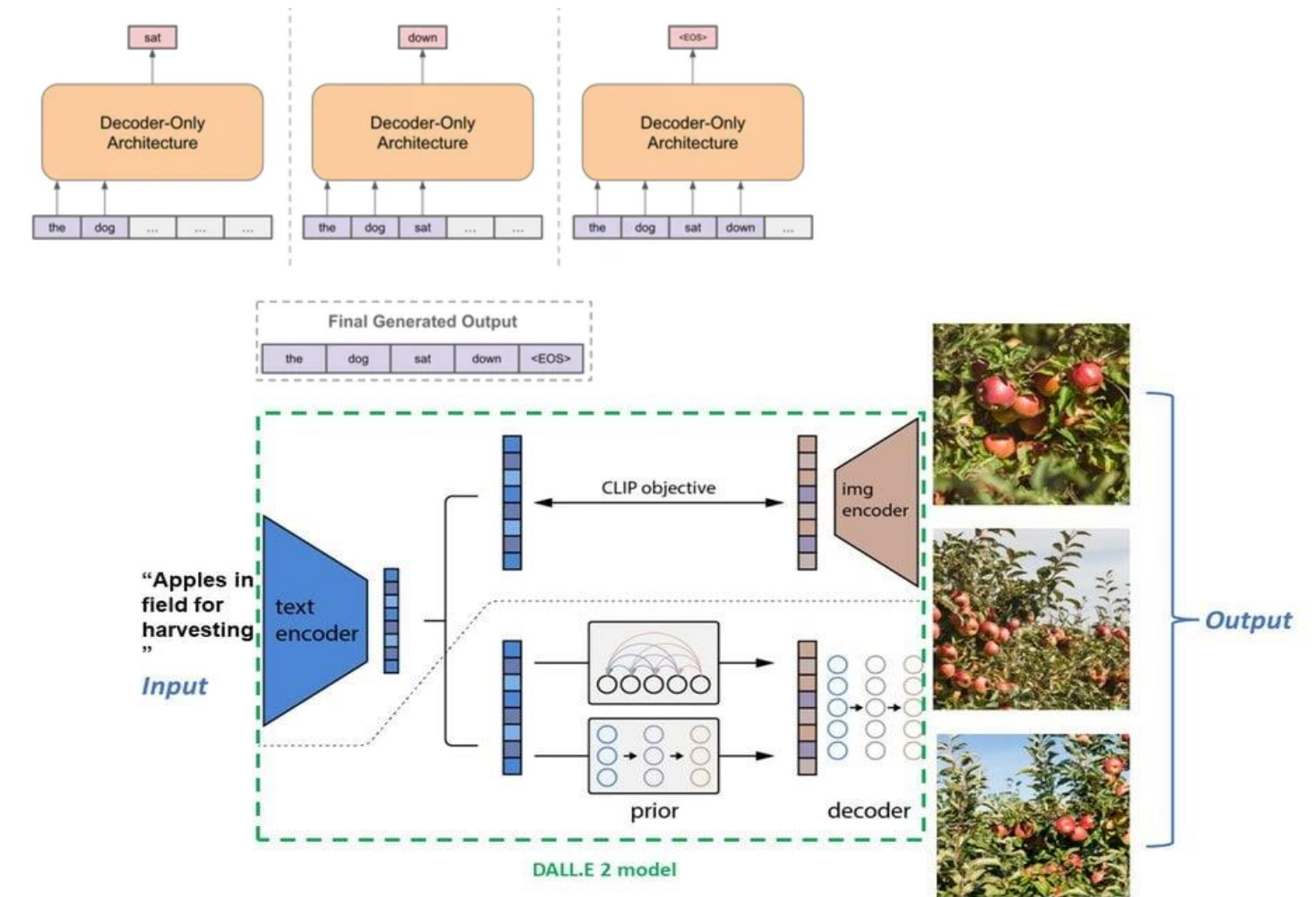
Embeddings

Neural Network Embedding of all books on Wikipedia (Koehrsen, 2018)

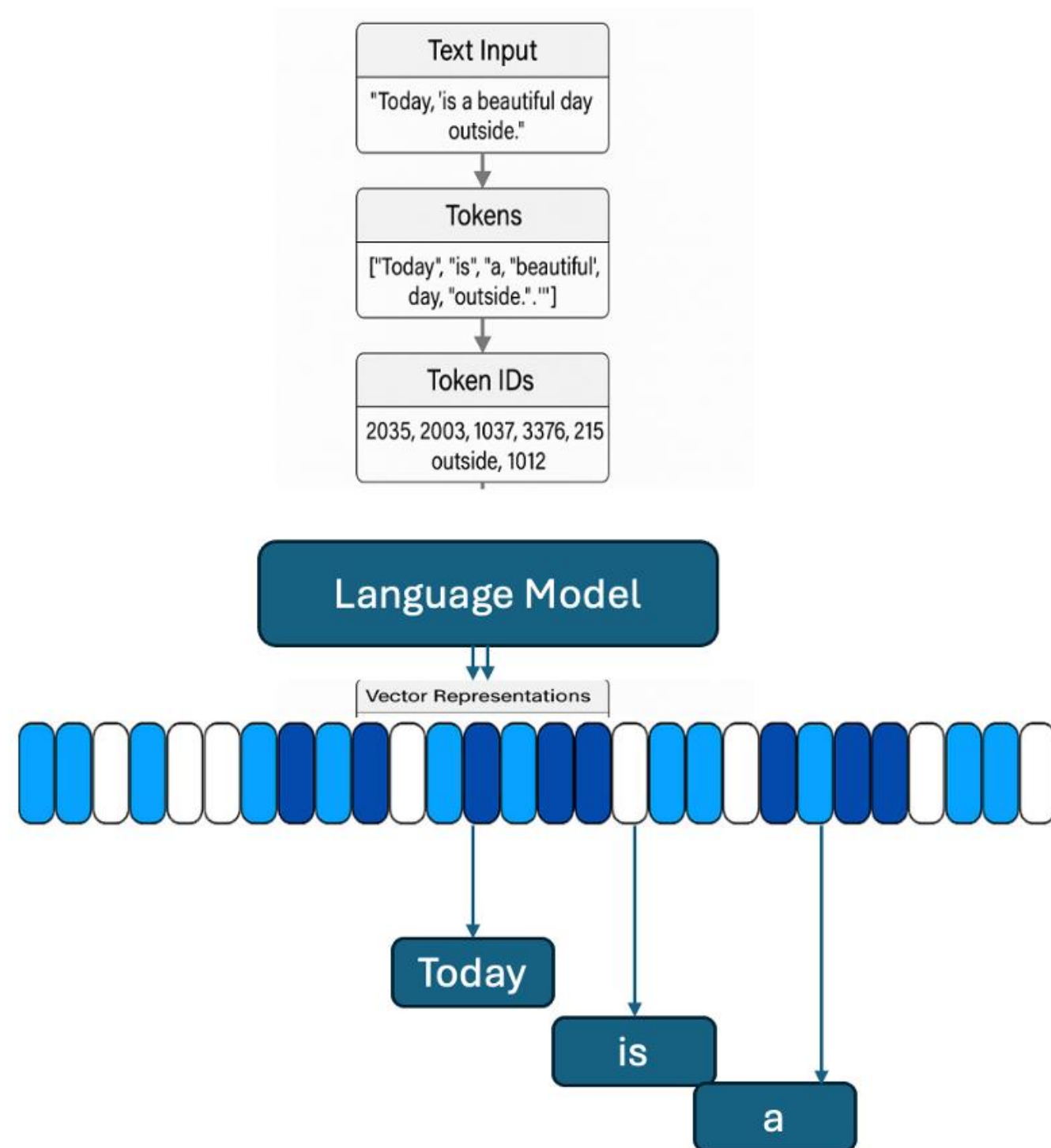


Generation Systems

- **Text generation:** GPTs, Llama, Claude – generate coherent content; predict the next token based on previous tokens (statistical likelihood)
- **Image generation:** DALL-E, Stable Diffusion, Imagen; generate images from prompts; Diffusion Models or GANS to create iteratively realistic images from random noise.
- **Audio generation:** Whisper (for ASR) or VALL-E (speech synthesis) learn representations of sound waves and either recognizes speech (ASR) or generates audio from text inputs.



Text Generation



- **Temperature:** Controls the randomness of the generation.
- **Top-k sampling:** Limit the next token choice to the top-k most probable tokens.
- **Top-p (nucleus) sample:** Select from the smallest set of tokens whose cumulative probability exceeds top-p
- These help regulate creativity vs precision.

Where do generation systems excel?

- Open-ended creativity tasks (storytelling, image creation, poetry, coding)
- Complex reasoning and problem-solving beyond retrieval capabilities
- Personalization and dynamic response generation (chatbots, educational tutors)
- Bridging gaps where no pre-existing data exactly fits the query

"Explain black holes to a 5-year-old using a cooking analogy."

"Imagine you're stirring a thick bowl of chocolate batter. If you spin it really fast, everything starts getting pulled toward the middle. A black hole is kind of like that in space—it pulls everything nearby toward it really strongly, even light, just like the batter getting pulled into the center of the bowl."

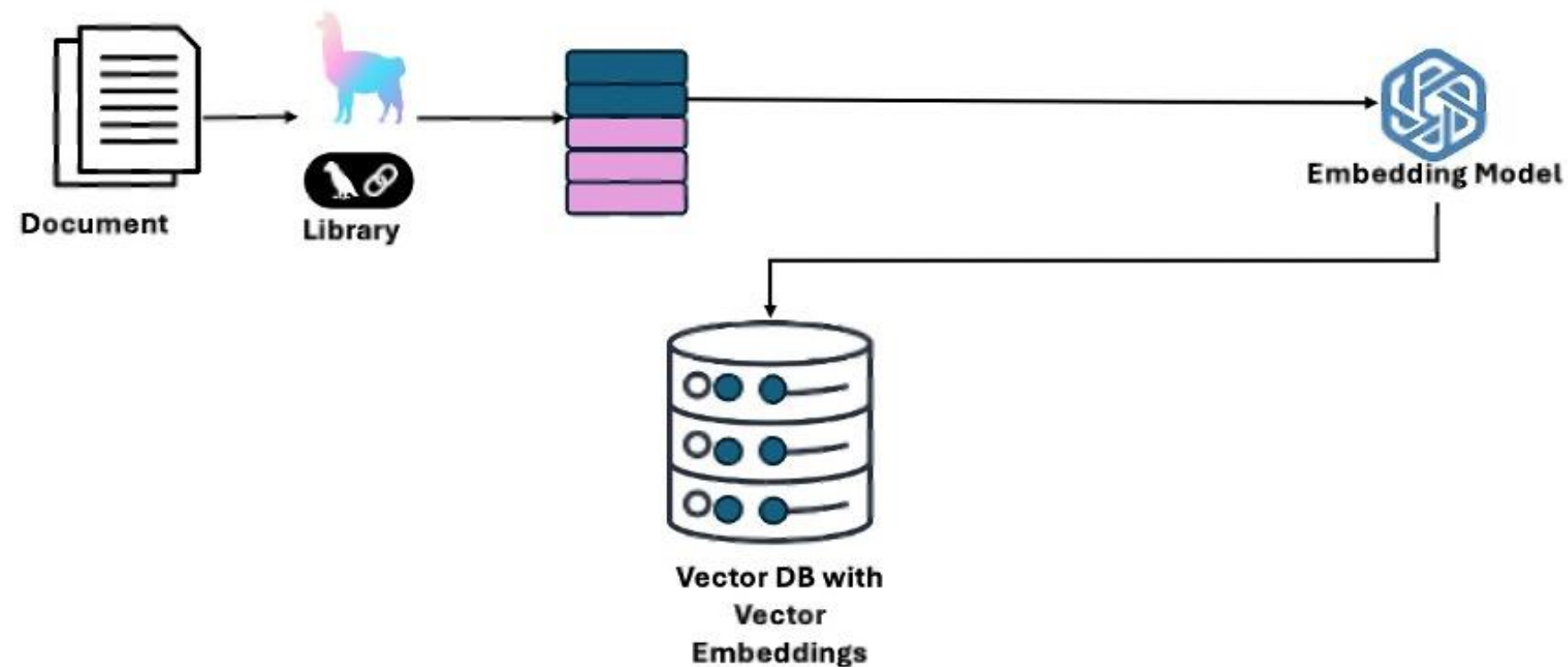
Retrieval – Augmented Generation

- Combines retrieval and generation into a unified workflow.
- Does not rely solely on the model's internal knowledge (may be outdated or incomplete)
- Two major steps:
 - **Retrieval Step:** Given a user query, the system first retrieves the top-k most relevant documents or chunks from an external knowledge base (eg. Vector database)
 - **Generation Step:** The retrieved documents are passed as context to an LLM, which generates the answer conditioned by the retrieved information.

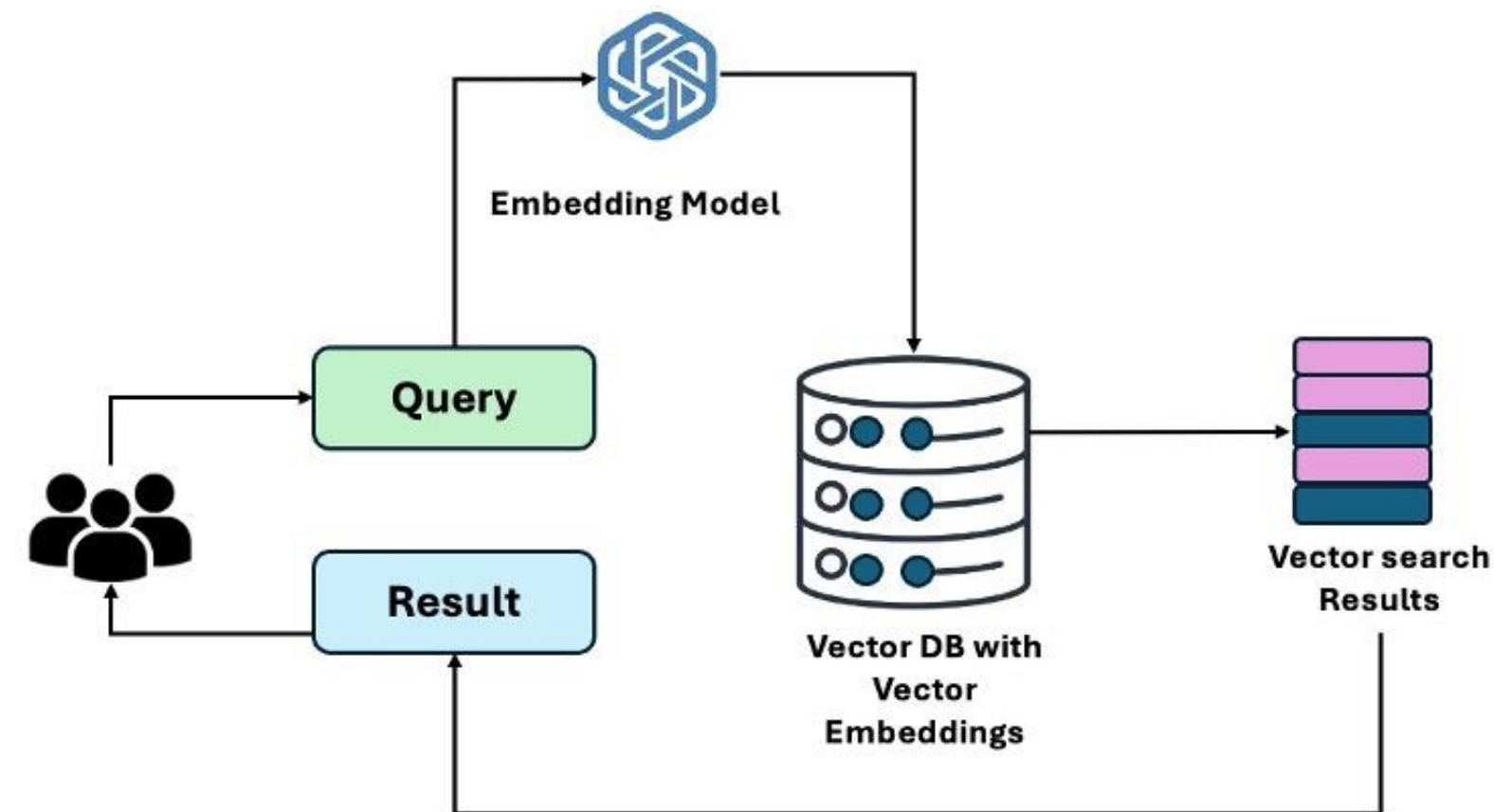
RAG Architectures

Single-Stage Architecture

- Simple pipeline: retrieve | generate
- Recommended when retrieval quality is high, and latency needs to be minimal



Offline indexing and embedding workflow

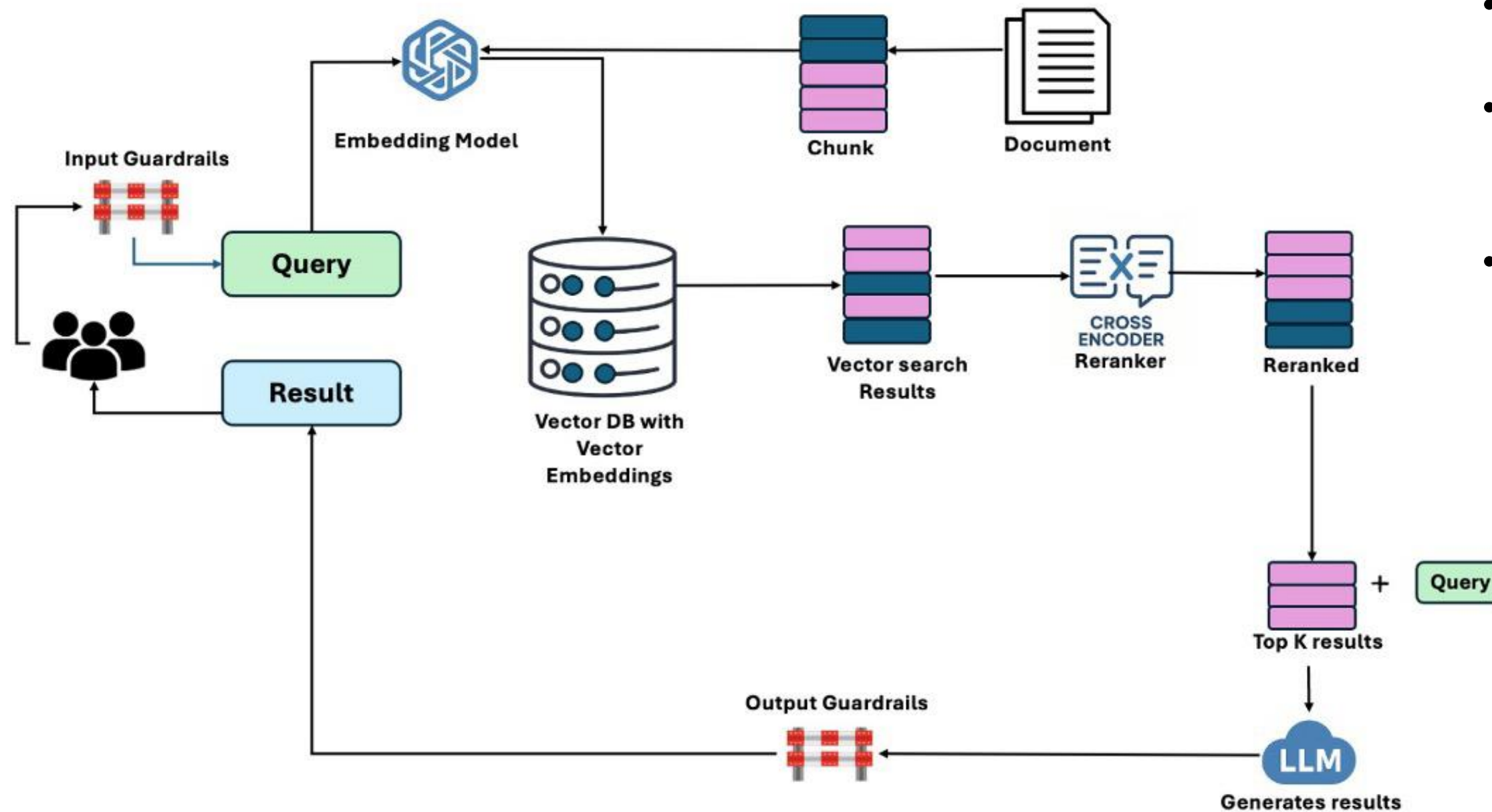


Querying the retrieval system

RAG Architectures

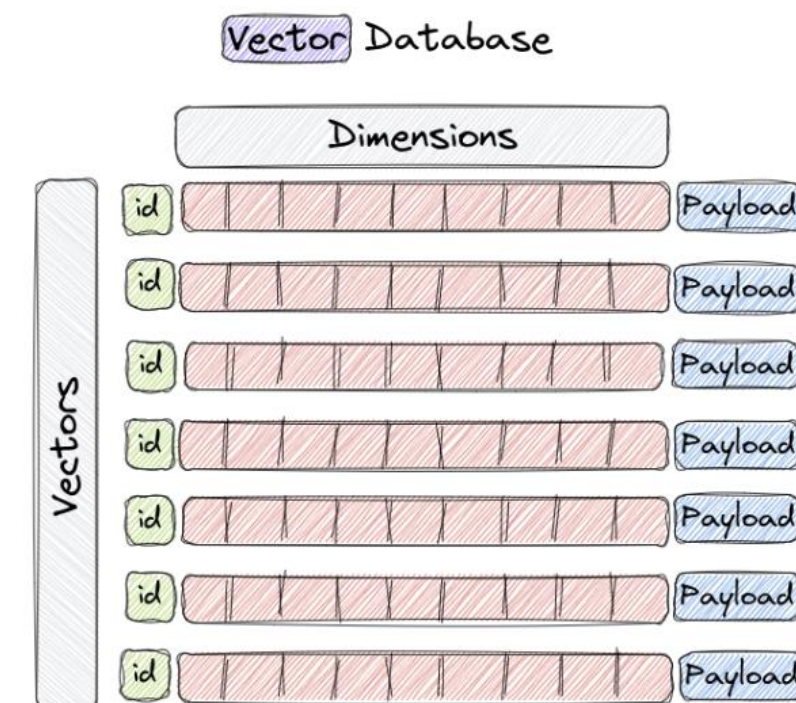
Two-Stage Architecture

- retrieval | reranking | generation
- A second model reranks the documents after the first retrieval to improve context quality
- Reduce hallucinations drastically.



Vector Database

- Designed to store and retrieve dense vector embeddings (numerical representations of text, images, and data)
- Search is based on distance metrics like Cosine Similarity or Euclidean Distance.
- The quality of retrieval depends heavily on the quality of embeddings.

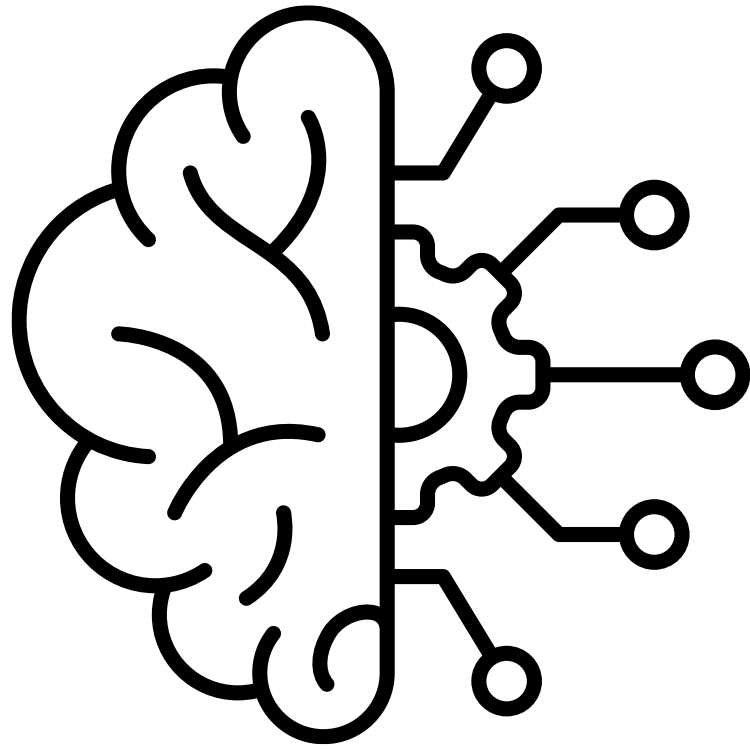


Guardrails

- Structured controls that guide and constrain model behaviour.
- Ensure the model acts safely, ethically, and in alignment with application or organizational goals.
- Reranking -> focuses on more relevant and factual information.
- Guardrails -> focuses on preventing bias propagation, policy violations, etc.
- Crucial factors in Agentic AI for healthcare, education, research, finance, etc.



Let's Practice on n^8n



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